CPSC 491- Senior Capstone Project in Computer Science

INSTRUCTOR:

Marc Velasco

CONTACT INFORMATION:

Email: mavelasco@fullerton.edu
Discord: https://discord.gg/Zdd4wRBvWr

- When you join make sure you drop your name and section number in General or via DM so I can add you to the proper role, each class has their own discord channel.

COURSE SCHEDULE

Class: M 4-6:45pm HUM409

OFFICE HOURS:

Office Hours: Mondays 15m before or after class, in our assigned classroom, by appt on Fridays

COURSE DESCRIPTION AND PURPOSE:

Per the University Catalog description:
A computer science research or real-world level of application development project. Business communication. Presenting the results to a wide range of audiences. Demonstrating the culminating experience of the practicum in computer science. Writing a final project report and technical documents such as user manuals, installation guides, feasibility study reports.

STUDENT LEARNING OBJECTIVES

- Apply computer science theories and principles, software development methods;
- Establish a software process;
- Work effectively in a team and communicate with a wide range of audiences;
- Complete a research or industry-level application project
- Write a high-quality project report; and
- Present and demonstrate the results to a wide-range of audiences.

Important: This syllabus is meant to reinforce and drive towards these objectives, but if for any reason I feel that the goals of the course are not being served based on the format of the class laid out here, I reserve the right to change, with notice given, the format, timing, any or any other part of the class and it's work in order to meet the learning objectives of the course.

CLASS FORMAT

To summarize and set expectations, this class will utilize groups for student work, each group will work on a project throughout the semester. Students will be evaluated individually on their **contribution** and **competency**.

- Groups will consist of between 2 and no more than 5 students, groups will be setup by me. If needed, I'll move students around to ensure proper number of students per group.
- Groups will have supervised control of tooling and technologies used in development while paying mind to the sprint schedule and adhering to any class requirements.
- All groups must be made and formed in the same section, **no** cross section/class groups.
- **No** individual projects or supervised work with another section/professor.
- All students are graded individually
- There will be **no** "incompletes" given out as grades, **no** makeup for missed meetings.

GENERATIVE AI TOOLS FOR DEVELOPMENT

Use of generative AI is part of this course. It is expected that you will make use of and show proof of interacting with tooling. You will be graded on the effective and efficient use of tooling to assist you but also graded on your ability to take responses from generative tools and formulate into best fit solutions to meet the needs of the class.

TEXTBOOKS AND RESOURCE MATERIALS:

- ChatGPT edu - https://www.fullerton.edu/it/services/software/chatgpt-edu.html, each student will need personal access and need to opt-in, you will utilize ChatGPT to construct and execute prompts and evaluate outputs for use in your project activities. Note I will ONLY allow you to utilize ChatGPT, part of your grade will be in having artifacts from ChatGPT to support your competency.
- Google docs and sheets will be used for artifacts such as plans and reports, test cases and results, as examples.
    - All edits must be performed with a named user account, same account throughout the semester).
- Code must be checked into github only
    - All checkins must be made with known user accounts (same account all throughout the semester), all code checkins must utilize pull requests and be peer reviewed.
- No textbook is required.
- Lecture notes and suggested reading materials will be posted on a learning management system

- Recommended textbooks: =
- *Docs for developers: an engineer's field guide to technical writing*
  Bhatti, Jared, author.; Hightower, Kelsey
  2021
- *Writing for engineering and science students: staking your claim*
  Rau, Gerald, 1954- author.
  2020
- A computer with a web browser and Internet access to the course website

COURSE GRADING:

Grades will be assigned as follows:

A+ 97–100%

A 93–96%

A− 90–92%

B+ 87–89%

B 83–86%

B− 80–82%

C+ 77–79%

C 73–76% Minimum for meeting the upper division Writing Requirement

C- 70–72%

D+ 67–69%

D 63–66%

D- 60–62%

F 0–59%

Note: While I make every effort to ensure accurate grading all work, canvas is not the official gradebook, make sure you are monitoring your grade and correct calculation of your weighted grades based on the table below, my final grades will be based on the weight below for all work.

I do round up to the nearest 1%, so 85.5 would round up to 86%, only if you have not missed any presentations or exceeded your group meeting absences.

WHAT CONSTITUTES YOUR GRADE:

Project Sprints	70%
Final Demo	10%
Final Project Report	10%
Midterm Demo and Project Report	10%

Note that all grading is individual, there is no comparison nor discussion on grading between students nor within the same group. Each grade is determined individually, and when discussing a grade there can be no discussion that involves other people (comparing grades etc.), I will only discuss a student's individual work unless there are allegations of academic dishonesty involved.

Also note the rubrics used are guidelines for performance, they are meant to show the example amount of effort and mastery needed for such type of grading, I reserve the right to use the rubrics as guidelines and record a grade that in my professional judgement reflects the effort and mastery exhibited in the work submitted. Such determinations fall under the guidance of [UPS300.030 Academic Grade Appeals](), where professional judgement of the instructor is relied upon for grading and assessment of work when it comes to course standards and course expectations.

PROJECT SPRINTS

Each sprint will consist of a set of artifacts (documents or content you develop) and results (code, tests executed, etc.) while using ChatGPT as an assistant.

Artifacts – these are identified by me at the beginning of a sprint, for example a project plan to detail what you will be doing and when, or a test plan to describe what testing will happen and when etc. You will turn in a draft during the first week of a sprint of this artifact.

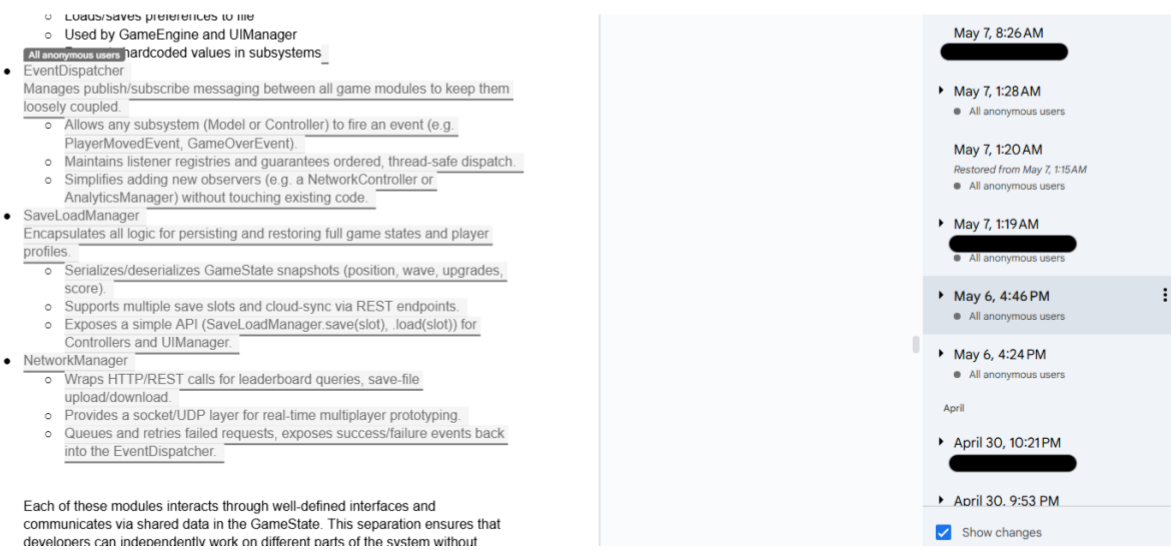
Results – What did you deliver during the sprint? This will mean code, tests, plans, processes, jira tracking your completed and assigned work. You can and should work on all this during class, however if you choose you can work on this part outside of class as well.

ChatGPT – record of prompts and saved chats, iterations of prompts and thoughts throughout the sprint.

Grading will be based on evidence of work in each of these areas. Minimal work or little effort will result in minimal or no points.

Anonymous work and/or lack of work, WILL NOT RECEIVE any points.

Example:



Note: You will earn points each sprint, you cannot make up points once they are gone. Each student is responsible for ensuring they are monitoring their progress and grades and if they find themselves in a position from which they cannot mathematically pass the class they should make use of all university tools/resources.

SPRINT DRAFTS

The first week of each sprint you will produce a draft of the artifact for that sprint for submission, this will not and should not be a complete artifact but merely a draft to show that you're utilizing class time to make progress.

Draft artifacts do not receive points, rather they are needed to ensure 100% possible points for that sprint, if you fail to submit a draft you will receive at most 70% for that sprint before grading occurs.

SPRINT PRESENTATIONS

Each week I will randomly call, at random times, a random number of students to present what they have, either a draft, or demo what they did if it's the end of a sprint.

If a student is not present or unable to present it will count the same as not being in class for that week and I will not count their work as submitted "in-class".

EXAMPLE RUBRICS FOR END OF A SPRINT

Sprint Artifacts	Exceeds	Mastery	Median	Near	No Evidence
view longer description	Excellent quality, clear mastery of concepts and topics demonstrated, clear evidence of discussing and evaluating multiple options and alternatives and justifying decisions. 35 pts	Moderate amounts of quality and writing, showing good grasp of concepts and goals for the project. 30 pts	Content developed with some elaboration, justification, and or demonstration of skill, but with room for improvement. Good examples of this are narrow scoped artifacts, for example a test plan that shows a single test technique with good definition and explanation, but limited in scope (forest from the trees view). 25 pts	Terse content with no elaboration of concepts or justification. Unfeasible content, or content simply pasted with no real adaption to make a best fit for the task at hand with given amount of resources. 15 pts	No artifacts submitted, or anonymous work submitted. 0 pts

ChatGPT Usage	Exceeds	Mastery	Wag the Dog
view longer description	Many separate instances of prompting showing iteration and discovery of new ideas, evolution of ideas into mature thoughts and actionable plans or paths. Able to take received output and convert to realistic plans and deliverables as evidenced in sprint turn-ins. 20 pts	Seferal instances of prompting showing attempts to incorporate responses into formulated ideas. Ideas not entirely feasible or adapted to task at hand. 10 pts	Unclear if a human was involved at all in any decision making, no adaption or effort to correct or reformulate anything that was generated, or no use of ChatGPT documented at all. 0 pts

Sprint Results	Exceeds	Mastery	Near	Below	No Evidence
view longer description	Excellent example of work accomplished, acquisition of new skills, and complexity of skill required for this sprint. Sustained work throughout the sprint (at least more than 4 days of work exhibited) 45 pts	Above average standard work and demonstration of skill, student shows ability to perform and execute on tasks assigned and deliver work of sufficient complexity and skill. (at least 3 separate days of work during the sprint) 32.14 pts	Able to deliver work of medium complexity, medium level of evidence of competency or skill required for work delivered, little sustained throughput (at most 2 separate days of work during the sprint) 25.71 pts	Minimal work done and evidenced, very low complexity or evidence of competency, at most 1 day of work or equivalent level of effort put into assigned tasks. 12.85 pts	No evidence of work or meaningful contributions evidenced throughout sprint. 0 pts

ATTENDANCE/SUBMISSION OF WORK/LATE POLICY

All work in this class will take place in person in the classroom via drafts, presentations and collaboration with the group.

All submissions must be done in class for 100% credit.

Work submitted outside of class (remote/after class has occurred) can only be done up unto the end of Sunday after the missed class and will result in a max possible grade for that sprint being 70% before grading occurs.

If no draft is ever submitted no work will be counted as submitted for that sprint at all.

If both weeks of a sprint are not submitted in class (or a presentation is missed or for some reason no verification of in class submission of work occurred) the maximum grade for the sprint you can get will be 50% before grading occurs.

Two consecutive sprints cannot be submitted remotely/not in class, if this is done, both sprints will be marked as 0 points total.

At the end of the term, I will throw away the single lowest sprint score among all your sprints.

PRACTICUM EXAM

If you somehow end up in a position where you are unable to pass the class and meet the upper division writing grade requirement of 73%, I will allow students who finish the class with a 65% or higher to take a closed book/device/notes **practicum exam** that will test your knowledge of CS theory and principles as it applies to the project you've delivered, if you score an 80% I will raise your grade to a 73%.

There will be no other makeup or alternative grade opportunities aside from this.

GROUP PRESENTATIONS

Group Project presentations will be required two times during the class. A student is responsible for personally presenting as part of a group, each student must speak live in front of the classroom and sharing via the room AV system, showing what they have worked on.

Every student MUST attend all designated presentations, even if you have given/presented already, all students are expected and graded on attending every single presentation from all groups.

If you are not in attendance for presentations:

- For Midterm presentations - student will receive a 25% penalty to your total points earned from all sprints up to the midterm.
- For Final presentations - student will receive a 30% penalty to your total points earned from between the midterm and the final presentation.

Students who are late to class on a presentation day will receive a penalty up to the maximum for a missed presentation depending on how late they are to the class session.

- 0-10 minutes late = 5% penalty
- 10-30 minutes late = 15% penalty
- 30+ minutes late = max penalty for missing a presentation (25% for midterm, 35% for final)

ACADEMIC DISHONESTY:

In regard to Academic Dishonesty please review the following link:
http://www.fullerton.edu/senate/publications_policies_resolutions/ups/UPS%20300/UPS%20300.021.pdf

DISABILITIES ACCOMMODATION:

If you have a disability that may qualify you for special accommodations in this course, you should contact: (**http://www.fullerton.edu/dss/**)

Office of Disabled Student Services
Contact: http://www.fullerton.edu/dss/contact_dss/index.php

TENTATIVE CLASS OUTLINE:

- First day – August 24th
- Sprint 1 – August 31-September 14th (Campus closed September 7th)
- **Sprint 2 – September 21st-28th**
- **Sprint 3 – October 5th-12th**
- **Midterm Demo and Presentation October 19th**
- **Sprint 4 – October 26th-November 2nd**
- **Sprint 5 – November 9th and 16th**
- **Fall Recess – November 23rd**
- **Sprint 6 – November 30th-December 7th**
- **Final Demo and Presentations – 12/14 5-6:50pm Hum 409**